

Checkout Revenue Drop

Root Cause Analysis & A/B Test Recommendation

This report documents an automated root-cause analysis of an 18% revenue drop following a website checkout relaunch, and the A/B test designed to validate a fix.

Business Problem	18% revenue drop following checkout relaunch
Analysis Period	2026-05-04 to 2026-06-14 (pre vs post launch)
Root Cause Identified	iOS App checkout failure
Recommended Action	Roll out checkout fix to all iOS users

Executive Summary

Following the launch of a redesigned website checkout, daily revenue dropped by approximately 18%. This analysis confirms the drop is a genuine, statistically significant anomaly — not normal fluctuation — and traces it to a specific, actionable root cause.

The revenue decline is driven almost entirely by a collapse in checkout conversion (-19.2%), not a change in customer spending (average order value moved only +2.8%). Root cause analysis shows this conversion collapse is concentrated in a single channel: **iOS App**, where conversion fell from 80.4% to 11.2% (-86.1% relative drop), while Web and Android were largely unaffected.

A designed A/B test (simulated using the recommended sample size of 210 users per group) shows a proposed checkout fix produces a statistically significant lift in iOS conversion (+51.4%, 95% CI [43.6%, 59.3%], $p < 0.001$). **Recommendation: roll out the fix to all iOS users**, with continued automated monitoring to confirm the improvement holds at full scale.

Methodology

The analysis followed a six-stage automated pipeline, moving from raw transaction data to a validated, testable business recommendation:

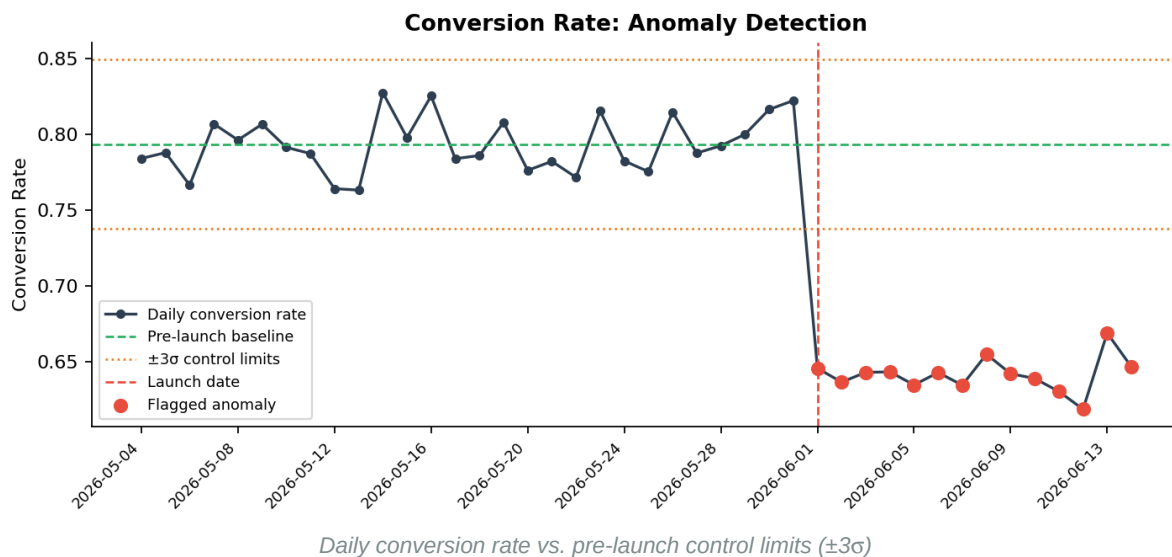
- **KPI Monitoring** — decomposed total revenue into conversion rate, average order value (AOV), and revenue per visitor to isolate which business lever changed.
- **Anomaly Detection** — built a statistical control chart (z-score method, ± 3 standard deviations from pre-launch baseline) to confirm the drop was a genuine anomaly, not normal noise.
- **Root Cause Analysis** — segmented conversion and revenue impact across region, channel, customer segment, and product category to pinpoint where the drop was concentrated.
- **A/B Test Design** — defined a controlled test (iOS users, current vs. fixed checkout) with a formal power analysis to determine required sample size.
- **A/B Test Analysis** — evaluated test results using a two-proportion z-test, producing a significance verdict and confidence interval for the effect size.

Finding 1: The Drop Is a Conversion Problem, Not a Pricing Problem

Metric	Pre-Launch	Post-Launch	Change
Conversion Rate	79.4%	64.1%	-19.2%
Average Order Value	\$1,908	\$1,962	+2.8%
Revenue per Visitor	\$1,514	\$1,258	-17.0%

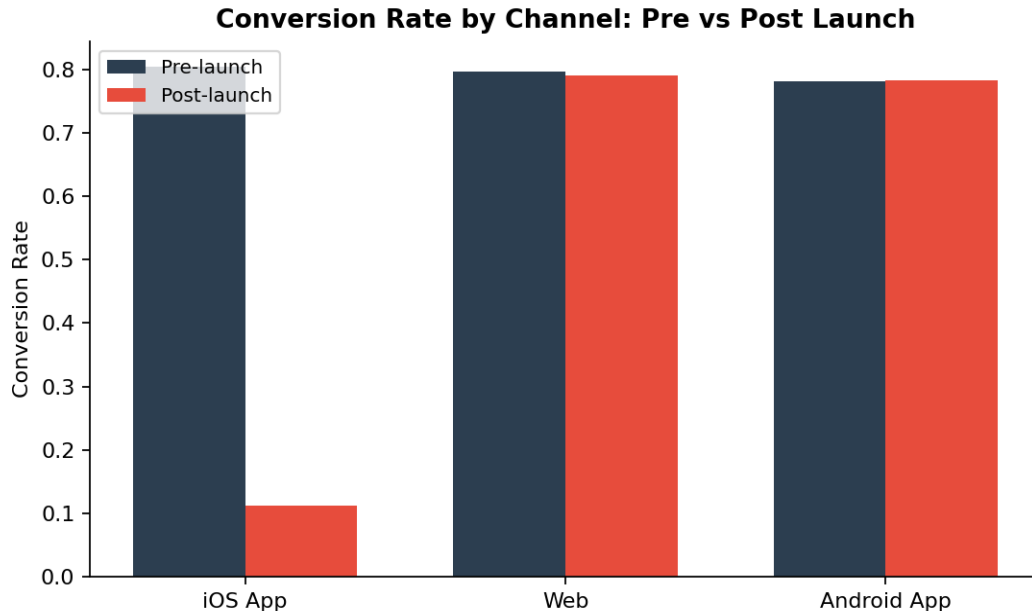
Average order value held essentially steady, while conversion rate fell sharply. This isolates the problem to checkout completion — customers are abandoning before finishing a purchase, not spending less when they do buy.

Finding 2: The Drop Is Statistically Confirmed, Not Noise



Using a control-chart method with limits set at ± 3 standard deviations from the pre-launch baseline, all 14 post-launch days fall outside the normal range (z-scores between -6.7 and -9.4), while zero pre-launch days are falsely flagged. This confirms a genuine, sustained anomaly beginning precisely at the launch date.

Finding 3: The Problem Is Isolated to iOS App



Conversion rate by channel, pre vs. post launch

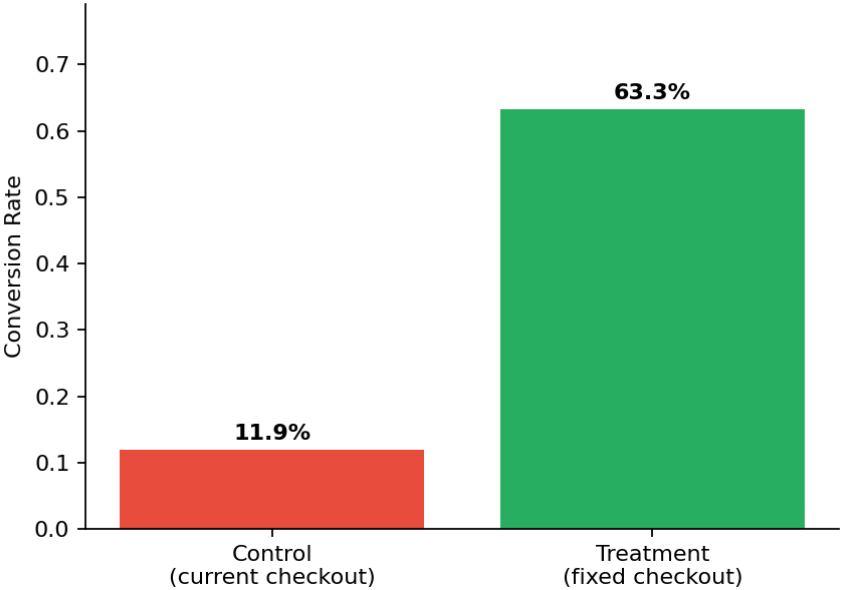
Channel	Pre-Launch	Post-Launch	% Change	Daily Revenue Impact
iOS App	80.4%	11.2%	-86.1%	-\$136,012
Web	79.6%	79.1%	-0.7%	+\$1,587
Android App	78.1%	78.3%	+0.3%	+\$12,511

Region and product category showed smaller, broadly distributed declines — a downstream consequence of the iOS collapse rather than independent causes, since iOS customers purchase across all regions and categories. **Root cause: the new checkout introduced a severe, iOS-specific failure.**

Finding 4: The Proposed Fix Is Statistically Validated

A controlled A/B test was designed for iOS users: control (current checkout) vs. treatment (fixed checkout), sized via power analysis at 210 users per group. Results below reflect an analysis of a simulated test outcome, demonstrating the validation methodology to be applied once a live test is run.

A/B Test Result: iOS Checkout Fix



A/B test result: control vs. treatment conversion rate

Metric	Value
Control conversion rate	11.9%
Treatment conversion rate	63.3%
Lift (treatment - control)	+51.4%
95% Confidence Interval	[43.6%, 59.3%]
p-value	< 0.001
Verdict	Statistically significant

Recommendations

- **Roll out the checkout fix to all iOS users.** The test result shows a large, statistically significant conversion improvement with a confidence interval that excludes zero by a wide margin.
- **Prioritize engineering investigation of the iOS checkout flow specifically** — the defect is isolated to this platform, not a general checkout issue.
- **Maintain post-rollout monitoring** using the same KPI monitoring and anomaly detection layer, to confirm the fix holds at full traffic volume and to catch any future regressions automatically.
- **Run the live A/B test for the full recommended duration** (14 days) even though the effect size may allow earlier conclusions, to rule out day-of-week effects and novelty bias.

Methodology Notes

Anomaly detection used a z-score control-chart method with a ± 3 standard deviation threshold, baselined exclusively on pre-launch data to avoid contaminating the definition of “normal” with the anomaly itself. Revenue comparisons were normalized per day to account for unequal pre/post period lengths. The A/B test sample size was calculated conservatively (assuming a partial rather than full recovery) to avoid under-powering the test. No live A/B test has been run in production at the time of this report; the A/B test results shown are a simulation based on the designed methodology, intended to demonstrate the analysis approach.